Lean & Six Sigma

Engineering Technology

May, 2026

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Lean & Six Sigma (A15602)

Short Title:	LSS		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Engineering		
Author	JLAWLOR		
Credits	5	**Level:**	Advanced

Description of Module / Aims

This module focuses on the tools and concepts associated with developing lean systems and achieving six sigma levels of excellence. Addressed at green belt level, this module examines the fundamentals of statistics, six sigma, and its application, as well as lean concepts and tools and their implementation. Related critical issues, including team performance, project management and voice of the customer, are examined in the context of achieving operational excellence.

Programmes

BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	4 / 8 / M
BSc (Hons) in Manufacturing Engineering (`WD_TCAMF_B`)	1 / 2 / M

Indicative Content

- Lean and six sigma---concepts and goals
- Basic probability, statistics, central limit theorem, distributions
- Collecting & summarising data, exploratory data analysis and hypothesis testing
- SPC, control charts, measurement system analysis, GR&R studies, Cpk, Ppk
- Design of experiments
- Lean process control, TPM, visual factory
- Project management basics from identification through closure, team performance
- Lean tools application, root cause analysis, problem solving, kaizen
- Management & planning tools, activity networks, PDPC, QFD

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Collect and summarise data, conduct hypothesis tests and interpret results.
2. Construct statistical process control charts, and apply a range of process capability indices.
3. Define basic DOE factors, and design DOE graphs and plots.
4. Apply lean tools to waste elimination, cycle time reduction and kaizen.
5. Utilise project planning tools, and evaluate the performance of project teams.
6. Plan and control processes through the implementation of TPM and visual factory.

Learning and Teaching Methods

- Lectures
- Presentations
- Guided reading

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1, 2, 4, 6
Continuous Assessment	30%	
In-Class Assessment	15%	3, 5
Assignment	15%	1, 2, 3, 4, 5, 6

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Montgomery, D.C. _Introduction to Statistical Quality Control_. 5th Ed.. Chichester: Wiley, 2005.
- Santos, J., R.A. Wysk and J.M. Torres. _Improving Production with Lean Thinking_. Chichester: Wiley, 2014.
- Various incl. Dombrowski, U. and T. Mielke. "Defining lean production: some conceptual and practical issues." _The TQM Journal_ 21, 2. (2009): 127-142.

Supplementary Material

- Barone, S. and E.L. Franco. _Statistical and Managerial Techniques for Six Sigma Methodology: Theory and Application_. Chichester: John Wiley & Sons, 2012.
- Bicheno, J. and M. Holweg. _The Lean Toolbox: The essential guide to lean transformation_. Buckingham: PICSIE Books, 2009.

- Henderson, G.R. _Six Sigma Quality Improvement with Minitab_. Chichester: John Wiley & Sons, 2011.
- Montgomery, D. _Design and Analysis of Experiments_. 6th Ed.. Chichester: Wiley, 2005.
- Taghizadegan, S. _Essential of Lean Six Sigma_. Oxford: Butterwirth-Heinemann, 2010.

Requested Resources

- Room Type: Computer Lab